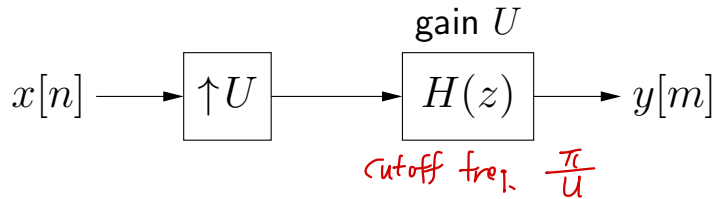


Multi-stage Interpolation:

- Recall the standard interpolation procedure:

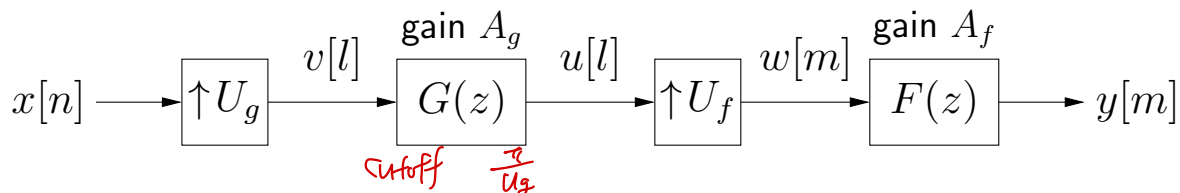


$$U = 10$$

$$U_g = 2 \quad U_f = 5$$

$$U_f = 5 \quad U_g = 2$$

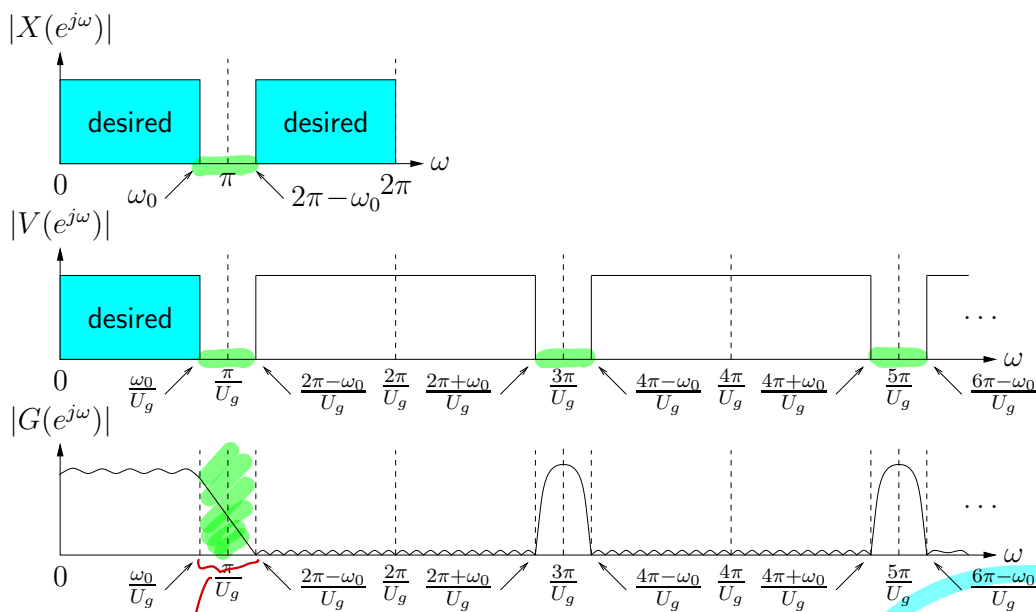
- It can be much more efficient to do this in two stages:



where $U_g U_f = U$ and $A_g A_f = U$. Let's see why...

- Assume $x[n]$ is bandlimited to $\omega_0 < \pi$ rad/samp. Then...

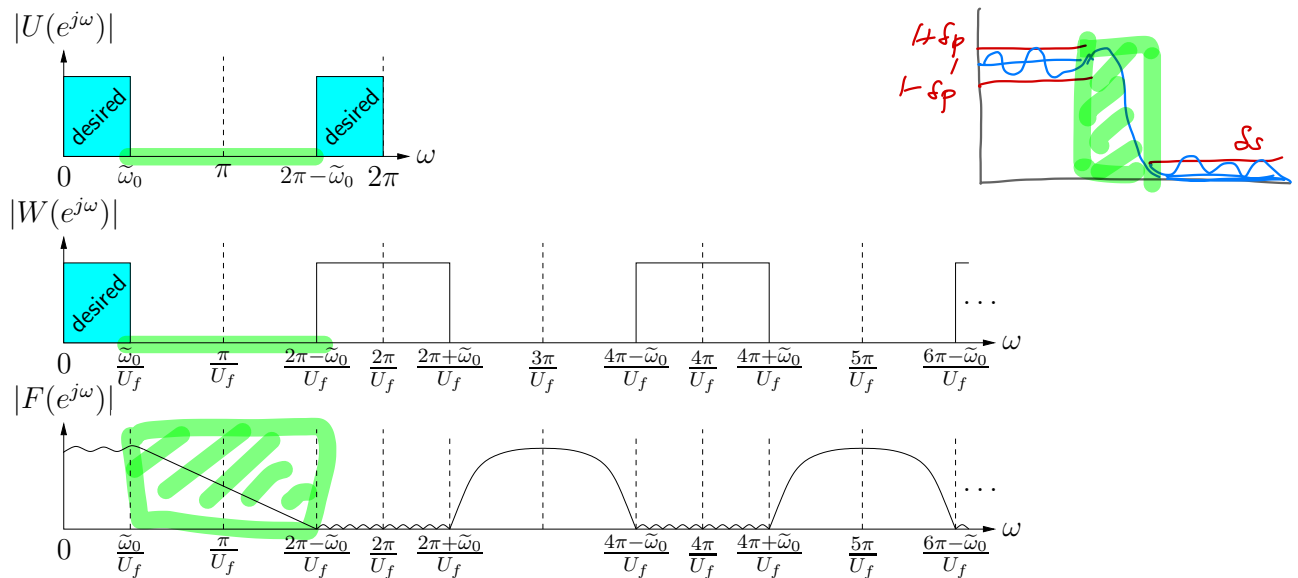
- $G(e^{j\omega})$ will have a passband cutoff at π/U_g and transition bandwidth $\Delta_g = \frac{2\pi - 2\omega_0}{U_g}$:



- Its output, $u[n]$, will be bandlimited to $\tilde{\omega}_0 \triangleq \frac{\omega_0}{U_g} < \frac{\pi}{2}$.

Multi-stage Interpolation (cont.):

- $F(e^{j\omega})$ will have a passband cutoff at π/U_f and transition bandwidth $\Delta_f = \frac{2\pi - 2\tilde{\omega}_0}{U_f} = \frac{2\pi U_g - 2\omega_0}{U}$.



- Say that δ_s and δ_p are defined relative to DC gain = 1. That is, we design unit-gain LPFs and then scale them.

- For the passband ripples, we want $U = A_g A_f$

$$U(1 + \delta_p) = A_g(1 + \delta_{p,g})A_f(1 + \delta_{p,f}) \approx U(1 + \delta_{p,g} + \delta_{p,f}) + \delta_{p,g}\delta_{p,f}$$

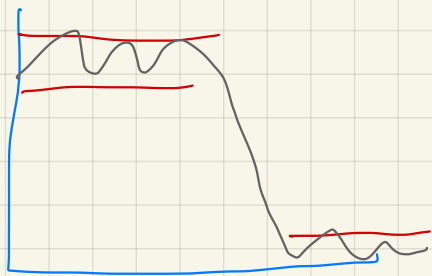
and so we set $\delta_{p,g} = \delta_p/2$ and $\delta_{p,f} = \delta_p/2$.

- For the stopband ripples, we want $A_g(1 + \delta_{p,g})A_f\delta_{s,f}$, $A_f(1 + \delta_{p,f})A_g\delta_{s,g}$, $A_f\delta_{s,f}A_g\delta_{s,g}$
- $$U\delta_s = \max \{ A_g \times A_f \delta_{s,f}, A_f \times A_g \delta_{s,g} \}$$

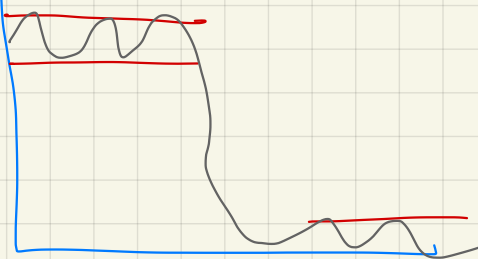
and so we set $\delta_{s,g} = \delta_s$ and $\delta_{s,f} = \delta_s$.

- How do we choose U_g and A_g (and thus U_f and A_f)?

$A_g(+\delta p_g)$



$A_f(+\delta p_f)$



X

Computational Savings of Multi-stage Interp:

- Kaiser's formula can be used to approximate the required filter lengths. For the single-stage $H(z)$ we have

$$L_h \approx \frac{-10 \log_{10}(\delta_p \delta_s) - 13}{2.3 \Delta_h} \triangleq \frac{\beta}{2.3(2\pi - 2\omega_0)} U$$

using $\Delta_h = \frac{2\pi - 2\omega_0}{U}$ and $\beta \triangleq -10 \log_{10}(\delta_s \delta_p) - 13$.

Thus L_h mults/input with polyphase implementation.

- For 2-stage interpolation, we have

$$L_g \approx \frac{-10 \log_{10}(\delta_s \delta_p / 2) - 13}{2.3 \Delta_g} \approx \frac{\beta + 3}{2.3(2\pi - 2\omega_0)} U_g$$
$$L_f \approx \frac{-10 \log_{10}(\delta_s \delta_p / 2) - 13}{2.3 \Delta_f} \approx \frac{\beta + 3}{2.3(2\pi U_g - 2\omega_0)} U$$

And with polyphase implementations of both stages, the total number of mults-per-input equals $L_g + U_g L_f$.

- Thus, the ratio of 2-stage to 1-stage multiplies equals

$$\frac{L_g + U_g L_f}{L_h} \approx \frac{\beta + 3}{\beta} \left(\frac{1}{U} + \frac{\pi - \omega_0}{\pi U_g - \omega_0} \right) U_g.$$

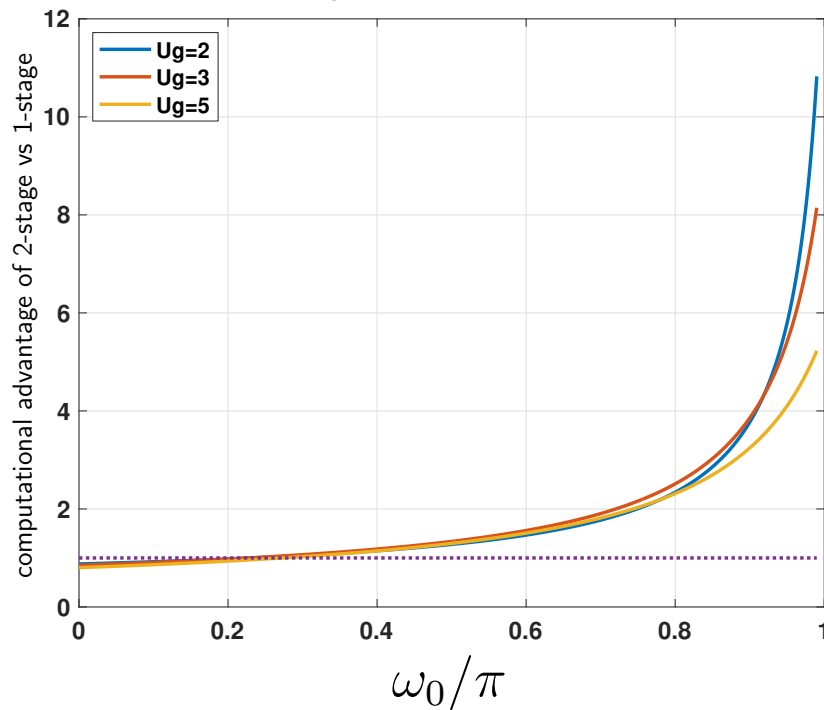
We can use this approximation to design U_g .

- Above eqns show that the choice of A_g doesn't matter.

$$30 = 2 \times 3 \times 5$$

Example of Multi-stage Interpolation:

- Suppose $U = 30$, $\delta_p = 0.002$, $\delta_s = 0.001$.
- The figure below shows the computational advantage of 2-stage over 1-stage. (Anything > 1 is an advantage.)

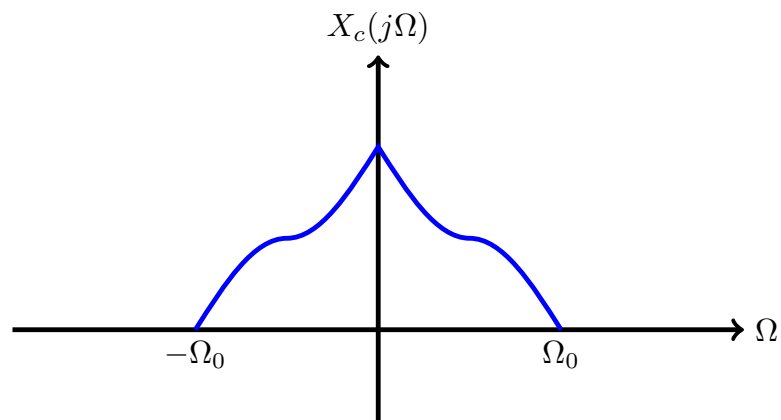


We see that...

- The advantage of using 2 stages grows as $\omega_0 \rightarrow \pi$.
- For large ω_0 , we should make U_g as small as possible.
- For medium or small ω_0 , choice of U_g doesn't matter.
- Above suggests little benefit to using 3 or more stages.
 - Little advantage to splitting 2nd stage because $\tilde{\omega}_0 < \frac{\pi}{2}$
- Similar ideas can be used for multi-stage decimation.

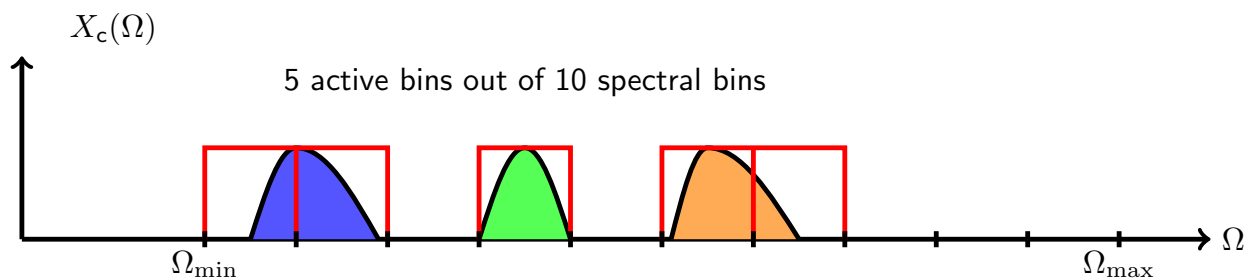
Sub-Nyquist Sampling of Multiband Signals:

- Conventional Nyquist sampling of bandlimited signals



$$x[n] = x_c(nT), \quad \frac{\pi}{T} \geq \Omega_0$$

- Multiband signal (still bandlimited but...)



- Q: Can we do better than Nyquist Sampling? How?

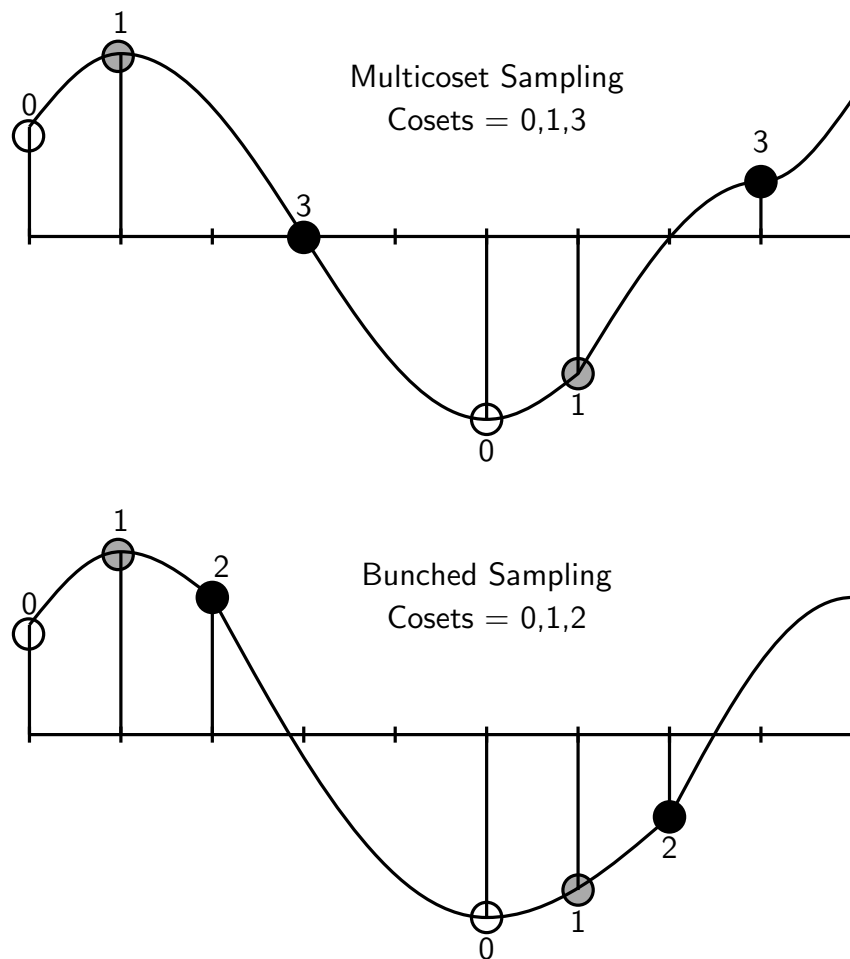
Multiset Sampling:

Let $\Phi = \{\phi_1, \phi_2 \cdots \phi_p\}$ be a subset of $\{0, 1, \dots, L-1\}$ of size $p < L$.

We acquire p discrete-time signals by

$$x_{\phi_k}[m] = x_c(LTm + \phi_k T), \quad k = 1, 2, \dots, p.$$

Example: $L = 4, p = 3,$

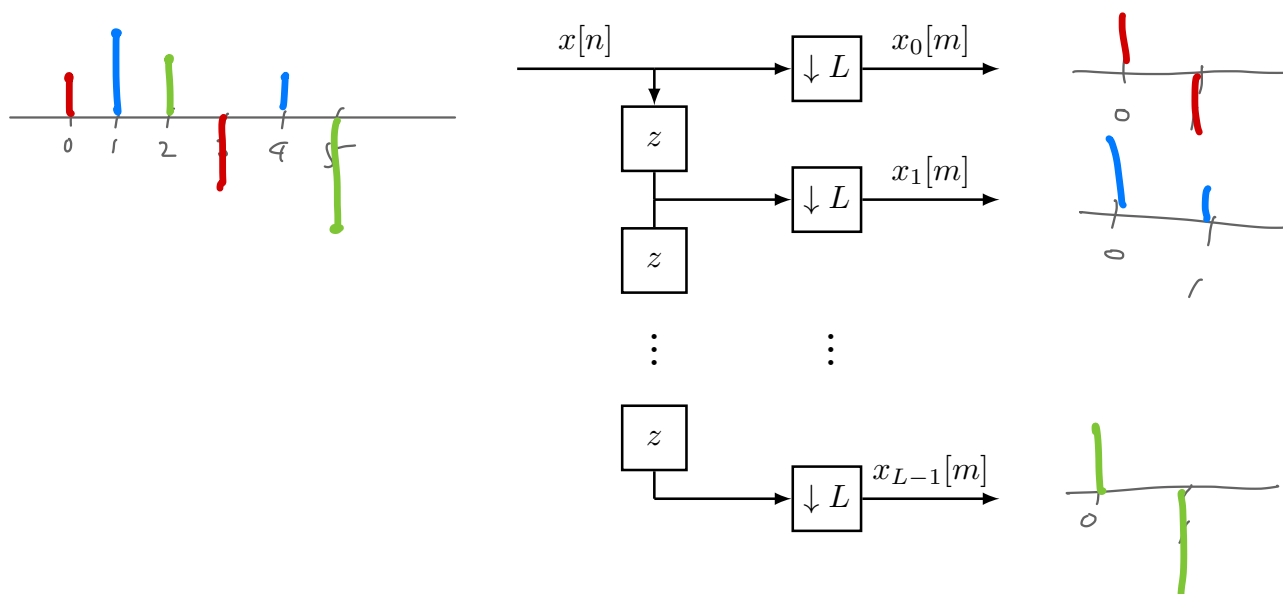


Polyphase Decomposition:

Polyphase components

$$x_k[m] = x[Lm + k], \quad k = 0, 1, \dots, L - 1$$

are obtained by the following block diagram:



Polyphase Decomposition of Nyquist-Rate Samples:

Let $(d_0, d_1, \dots, d_{L-1})$ denote the selection sequence such that

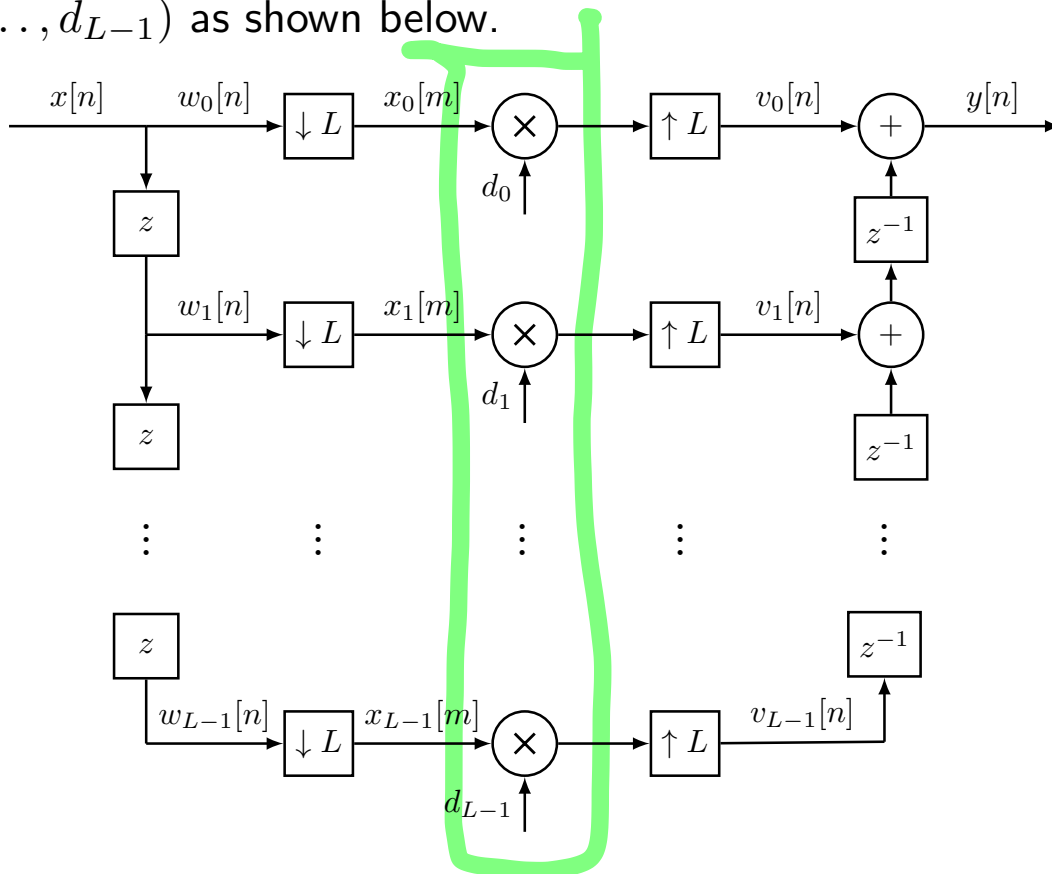
$$k \in \Phi \implies d_k = 1, \quad k \notin \Phi \implies d_k = 0.$$

Example: $L = 4, p = 3$

$$\Phi = \{0, 1, 3\} \iff \begin{bmatrix} d_0 \\ d_1 \\ d_2 \\ d_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

*index set
of selected components*

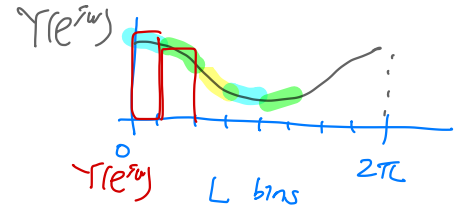
We consider a subset of polyphase components with respect to $(d_0, d_1, \dots, d_{L-1})$ as shown below.



$d_k = 1 \rightarrow$ select k th component / $d_k = 0 \rightarrow$ discard k th component

The output is written in the time domain as

$$y[n] = \sum_{k=0}^{L-1} v_k[n - k].$$



Therefore, we have

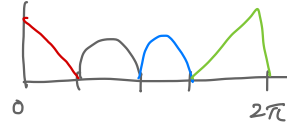
$$Y(e^{j\omega}) = \sum_{k=0}^{L-1} V_k(e^{j\omega}) e^{-j\omega k} = \sum_{k=0}^{L-1} d_k X_k(e^{j\omega L}) e^{-j\omega k}.$$

upsampling

By substituting ω by $\omega + \frac{2\pi i}{L}$, we obtain

$$Y(e^{j(\omega + \frac{2\pi i}{L})}) = \sum_{k=0}^{L-1} d_k X_k(e^{j\omega L}) e^{-j\omega k} e^{-\frac{j2\pi i k}{L}}, \quad i = 0, 1, \dots, L-1.$$

for $0 \leq \omega < \frac{2\pi}{L}$



The set of identities is rearranged to

$$\underbrace{\begin{bmatrix} Y(e^{j\omega}) \\ Y(e^{j(\omega + \frac{2\pi}{L})}) \\ \vdots \\ Y(e^{j(\omega + \frac{2\pi(L-1)}{L})}) \end{bmatrix}}_{\gamma(\omega)} = \mathbf{W} \underbrace{\begin{bmatrix} d_0 e^{-j0} X_0(e^{j\omega L}) \\ d_1 e^{-j\omega} X_1(e^{j\omega L}) \\ \vdots \\ d_{L-1} e^{-j\omega(L-1)} X_{L-1}(e^{j\omega L}) \end{bmatrix}}_{\alpha(\omega)},$$

where $\mathbf{W} \in \mathbb{C}^{L \times L}$ denotes the DFT matrix of size L . Let $\gamma_i(\omega)$ denotes the i th entry of $\gamma(\omega)$ for $i = 0, 1, \dots, L-1$, that is,

$$\gamma(\omega) = \begin{bmatrix} \gamma_0(\omega) \\ \gamma_1(\omega) \\ \vdots \\ \gamma_{L-1}(\omega) \end{bmatrix}.$$